Useful Formulas

Determinant of a 2x2 matrix

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}$$

Determinant of a 3x3 matrix

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} a_{11} - \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} a_{12} + \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} a_{13}$$

Mean and Standard Deviation of N Samples

$$\text{Mean: } \mu = \frac{\sum_{i=1}^{N} v_i}{N} \qquad \text{Standard Deviation: } \sigma = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(v_i - \mu)^2}$$

Branches/Loop/Networks: $b = l + n - 1$

Ohm's Law: $V = IR$

Series Resistance: $R_s = \sum_{i=1}^{N} R_i$

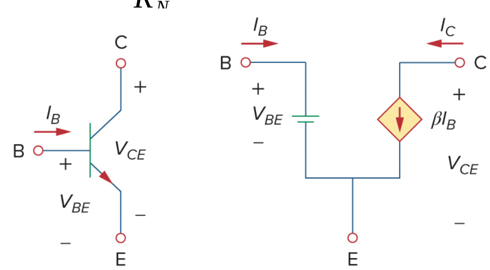
Parallel Resistance: $R_p = \dfrac{1}{\sum_{i=1}^{N} \dfrac{1}{R_i}}$

Kirchoff's Voltage Law (KVL): $\sum_{i=1}^{N} V_i = 0$

Kirchoff's Current Law (KCL): $\sum_{i=1}^{N} i_i = 0$

Norton Equivalent Source: $I_N = \dfrac{V_{TH}}{R_N}, \text{where } R_N = R_{TH}$

Transistor Equivalent Model

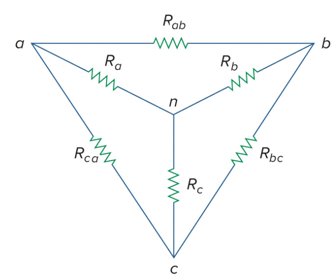


Delta-Wye Transformation

$$R_a = \frac{R_{ca}R_{ab}}{R_{ab}+R_{bc}+R_{ca}} \qquad R_{ab} = \frac{R_aR_b+R_bR_c+R_cR_a}{R_c}$$

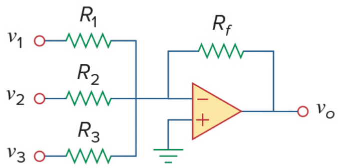
$$R_b = \frac{R_{ab}R_{bc}}{R_{ab}+R_{bc}+R_{ca}} \qquad R_{bc} = \frac{R_aR_b+R_bR_c+R_cR_a}{R_a}$$

$$R_c = \frac{R_{bc}R_{ca}}{R_{ab}+R_{bc}+R_{ca}} \qquad R_{ca} = \frac{R_aR_b+R_bR_c+R_cR_a}{R_b}$$



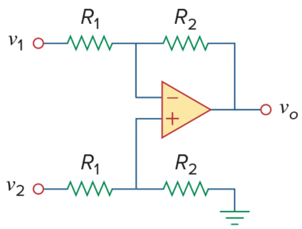
Useful Formulas

Op-Amp Configurations



Summer

$$v_o = -\left(\frac{R_f}{R_1} v_1 + \frac{R_f}{R_2} v_2 + \frac{R_f}{R_3} v_3\right)$$



Difference amplifier

$$v_o = \frac{R_2}{R_1}(v_2 - v_1)$$

Passive Elements Table

Relation	Resistor (R)	Capacitor (C)	Inductor (L)
v-i:	$v = iR$	$v = \frac{1}{C}\int_{t_0}^{t} i(\tau)\,d\tau + v(t_0)$	$v = L\frac{di}{dt}$
i-v:	$i = v/R$	$i = C\frac{dv}{dt}$	$i = \frac{1}{L}\int_{t_0}^{t} v(\tau)\,d\tau + i(t_0)$
p or w:	$p = i^2R = \frac{v^2}{R}$	$w = \frac{1}{2}Cv^2$	$w = \frac{1}{2}Li^2$
Series:	$R_{eq} = R_1 + R_2$	$C_{eq} = \frac{C_1C_2}{C_1 + C_2}$	$L_{eq} = L_1 + L_2$
Parallel:	$R_{eq} = \frac{R_1R_2}{R_1 + R_2}$	$C_{eq} = C_1 + C_2$	$L_{eq} = \frac{L_1L_2}{L_1 + L_2}$
At dc:	Same	Open circuit	Short circuit
Circuit variable that cannot change abruptly:	Not applicable	v	i

1st Order Circuits

$$x(t) = x(\infty) + [x(0) - x(\infty)]e^{-\frac{t}{\tau}}$$

$$\tau = RC \text{ or } \tau = \frac{L}{R}$$

2nd Order Circuits

$$\alpha = \frac{R}{2L} \text{ (series) or } \alpha = \frac{1}{2RC} \text{ (parallel)}$$

$$\omega_0 = \frac{1}{\sqrt{LC}} \qquad \omega_d = \sqrt{\omega_0^2 - \alpha^2} \qquad s = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$

Over-damped: $x(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$

Critically Damped: $x(t) = (A_2 + A_1 t)e^{-\alpha t}$

Under-damped: $x(t) = e^{-\alpha t}[B_1 \cos(\omega_d t) + B_2 \sin(\omega_d t)]$

AC Power Analysis

Complex Power $= S = P + jQ = V_{rms}(I_{rms})^* = |V_{rms}||I_{rms}|\angle(\theta_v - \theta_i)$

Apparent Power $= S = |S| = |V_{rms}||I_{rms}| = \sqrt{P^2 + Q^2}$

Real Power $= P = \mathrm{Re}(S) = S\cos(\theta_v - \theta_i)$

Reactive Power $= Q = \mathrm{Im}(S) = S\sin(\theta_v - \theta_i)$

Power Factor $= \frac{P}{S} = \cos(\theta_v - \theta_i)$

AC Phase to Line Voltage

$V_{an} = V_p \measuredangle 0^o \quad V_{bn} = V_p \measuredangle -120^o \quad V_{bn} = V_p \measuredangle 120^o$ (positive sequence)

$V_{ab} = \sqrt{3}V_p \measuredangle 30^o \quad V_{bc} = \sqrt{3}V_p \measuredangle -90^o \quad V_{ca} = \sqrt{3}V_p \measuredangle -210^o$

Transfer Function (Standard Form)

$$H(\omega) = \frac{K(j\omega)^{\pm 1}(1 + j\omega/z_1)[1 + j2\zeta_1\omega/\omega_k + (j\omega/\omega_k)^2]}{(1 + j\omega/p_1)[1 + j2\zeta_2\omega/\omega_n + (j\omega/\omega_n)^2]}$$

Laplace Transforms

Fourier Series	$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\frac{n\pi x}{p} + b_n \sin\frac{n\pi x}{p}$ $a_0 = \frac{1}{p}\int_{-p}^{p} f(x)dx$ $a_n = \frac{1}{p}\int_{-p}^{p} f(x)\cos\frac{n\pi x}{p}dx$ $b_n = \frac{1}{p}\int_{-p}^{p} f(x)\sin\frac{n\pi x}{p}dx$
Sine Series	$f(x) \sim \sum_{n=1}^{\infty} b_n \sin\frac{n\pi x}{P}$ $b_n = \frac{2}{p}\int_{0}^{p} f(x)\sin\frac{n\pi x}{p}dx$
Cosine Series	$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\frac{n\pi x}{p}$ $a_0 = \frac{2}{p}\int_{0}^{p} f(x)dx$ $a_n = \frac{2}{p}\int_{0}^{p} f(x)\cos\frac{n\pi x}{p}dx$
An Important Trigonometric Identity	$\cos(a+b) = \cos(a)\cos(b) - \sin(a)\sin(b)$ $\sin(a+b) = \sin(a)\cos(b) + \cos(a)\sin(b)$ $c_1\sin(\omega x) + c_2\cos(\omega x) = A\sin(\omega x + \phi)$ $A = \sqrt{c_1^2 + c_2^2}$ $\phi = \tan^{-1}\left(\frac{c_2}{c_1}\right)$ **Note**: **(1)** to calculate ϕ properly, c_1 must be the constant assigned the $\sin(\omega x)$ and c_2 must be the constant assigned the $\cos(\omega x)$. **(2)** If $c_1 < 0$, add π to the answer provided by your calculator.

Laplace Transforms

	$f(t)$	$F(s) = L\{f(t)\}$
Definition	$f(t)$	$\int_0^\infty e^{-st} f(t)dt$
Basic Forms	1	$\frac{1}{s}$
	t^n	$\frac{n!}{s^{n+1}}$
	e^{at}	$\frac{1}{s-a}$
	$\sin(kt)$	$\frac{k}{s^2+k^2}$
	$\cos(kt)$	$\frac{s}{s^2+k^2}$
Derivative Forms	$f'(t)$	$sF(s) - f(0)$
	$f''(t)$	$s^2F(s) - sf(0) - f'(0)$
	$f'''(t)$	$s^3F(s) - s^2f(0) - sf'(0) - f''(0)$
	$f^{(n)}(t)$	$s^nF(s) - s^{n-1}f(0) - \ldots - f^{n-1}(0)$
Translation	$e^{at}f(t)$	$F(s-a)$
	$e^{at}\sin(kt)$	$\frac{k}{(s-a)^2+k^2}$
	$e^{at}\cos(kt)$	$\frac{s-a}{(s-a)^2+k^2}$
Unit Step Functions	$U(t-a)$	$\frac{e^{-as}}{s}$
	$f(t-a)U(t-a)$	$e^{-as}F(s)$
	$f(t)U(t-a)$	$e^{-as}L\{f(t+a)\}$
Derivatives of Transforms	$t^n f(t)$	$(-1)^n \frac{d^n}{ds^n}F(s)$
Convolution	$f * g$	$F(s)G(s)$
	Note: $f * g = \int_0^t f(\tau)g(t-\tau)d\tau$	
	$\int_0^t f(\tau)d\tau$	$\frac{F(s)}{s}$
Dirac Delta	$\delta(t-a)$	e^{-as}